

BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```

Activating project at `~/Downloads/Cornell/BEE 5750/hw2-jinry/hw2-jinry`
Precompiling project...
 5678.6 ms    Plots → UnitfulExt
25084.2 ms    DataFrames
 1370.2 ms    Latexify → DataFramesExt
 3 dependencies successfully precompiled in 28 seconds. 252 already precompiled.

```

```

using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots

```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

```

using Pkg
Pkg.add("GraphRecipes")

```

```

Updating registry at `~/.julia/registries/General.toml`
Resolving package versions...
No Changes to `~/Downloads/Cornell/BEE 5750/hw2-jinry/hw2-jinry/Project.toml`
No Changes to `~/Downloads/Cornell/BEE 5750/hw2-jinry/hw2-jinry/Manifest.toml`

```

```

using GraphRecipes, Plots

A = [0 1 0;
      0 0 1;
      1 0 0]

names = ["Atmospheric CO ", "Global Temperature", "Ocean CO "]

edge_labels = Dict{Tuple{Int, Int}, String}((1, 2) => "+", (2, 3) => "+", (3, 1) => "+")

```

```

shapes = [:rect, :rect, :rect]
xpos = [0, -1, 1]
ypos = [1, 0, 0]

p = graphplot(A, names=names, edgelabel=edge_labels, markersize=0.07,
              markershapes=shapes, markercolor=:lightblue, x=xpos, y=ypos,
              arrow=:true, curves=false, linewidth=2, fontsize=8,
              title="CO Cycle Feedback System Diagram")

x_start, y_start = xpos[1] + 0.15, ypos[1] - 0.1
x_end, y_end = xpos[3] - 0.09, ypos[3] + 0.13

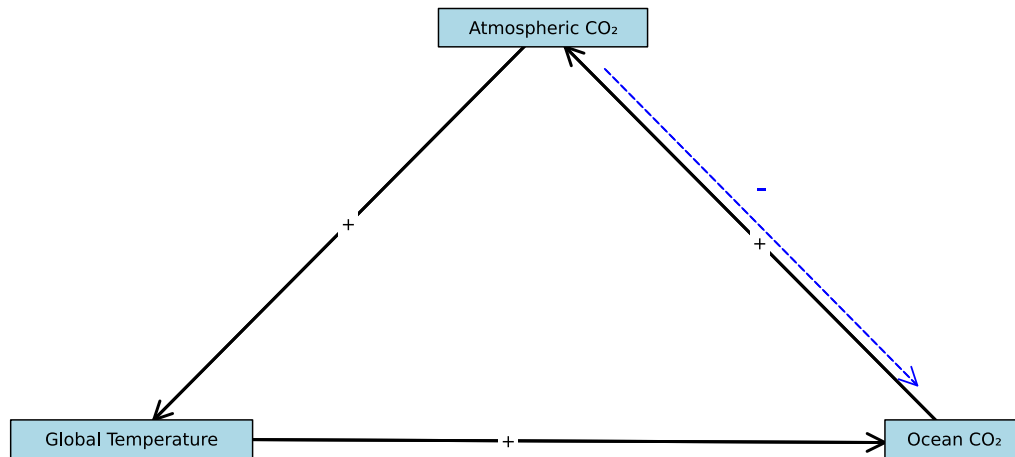
plot!([x_start, x_end], [y_start, y_end],
      line=:dash, linewidth=1.2, color=:blue, arrow=:true, arrowhead=:triangle, size=(800, 500))

annotate!((x_start + x_end)/2, (y_start + y_end)/2 + 0.1,
          text("-", 15, :blue, :center))

display(p)

```

CO₂ Cycle Feedback System Diagram



I drew it base on the understanding that when the concentration of atmospheric CO₂ rises, it acts as a greenhouse gas, which leads to an increase in the global mean temperature. This rise in temperature then negatively affects the ocean's ability to store CO₂, mainly by reducing the solubility of CO₂ in seawater. Consequently, the ocean releases more CO₂ back into the atmosphere, reinforcing this cycle. The dashed arrow is drawn because of Henry's Law. When air has more CO₂ than the ocean, the gas naturally dissolves into the seawater. This helps to lower the CO₂ in the air.

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

```
using GraphRecipes, Plots

A = [0 1;
     0 0]

names = ["Volume 1\n(0-15 km)",
        "Volume 2\n(15+ km)"]

edge_labels = Dict{Tuple{Int, Int}}{String}((1, 2) => "River Flow\n10 km/d")

shapes = [:rect, :rect]
xpos = [0, 8]
ypos = [0, 0]

p = graphplot(A,
              names=names,
              edgelabel=edge_labels,
              markersize=0.3,
              markershapes=shapes,
              markercolor=:lightblue,
              x=xpos, y=ypos,
              arrow=true, curves=false, linewidth=5, fontsize=10,
              title="River CRUD Transport System Diagram",
              xlim=(-3, 13), ylim=(-3, 5),
              size=(1100, 500))

plot!([-5.25, -1.2], [0, 0], line=:solid, linewidth=2, color=:blue, arrow=true, arrowhead=:triangleleft)
annotate!(-3, 0.5, text("Inflow\nQ=250,000 m³/d\nC=0.5 kg/1000m³", 9, :blue, :center))

plot!([9.25, 11], [0, 0], line=:solid, linewidth=2, color=:blue, arrow=true, arrowhead=:triangleleft)
annotate!(11.5, 0.5, text("Downstream\nOutflow", 9, :blue, :center))

plot!([0, 0], [2.5, 0.3], line=:dash, linewidth=2, color=:brown, arrow=true, arrowhead=:triangleleft)
annotate!(0.3, 2.8, text("Wastewater 1\nQ=40,000 m³/d\nC=9 kg/1000m³", 8, :brown, :left))

plot!([8, 8], [2.5, 0.3], line=:dash, linewidth=2, color=:brown, arrow=true, arrowhead=:triangleleft)
```

```

annotate!(8.3, 2.8, text("Wastewater 2\nQ=60,000 m3/d\nC=7 kg/1000m3", 8, :brown, :left))

plot!([2, 2], [-2, -0.7], line=:dash, linewidth=2, color=:green, arrow=true, arrowhead=:triangleup)
annotate!(2, -2.3, text("Deposition\n54 kg/km/d", 8, :green, :center))

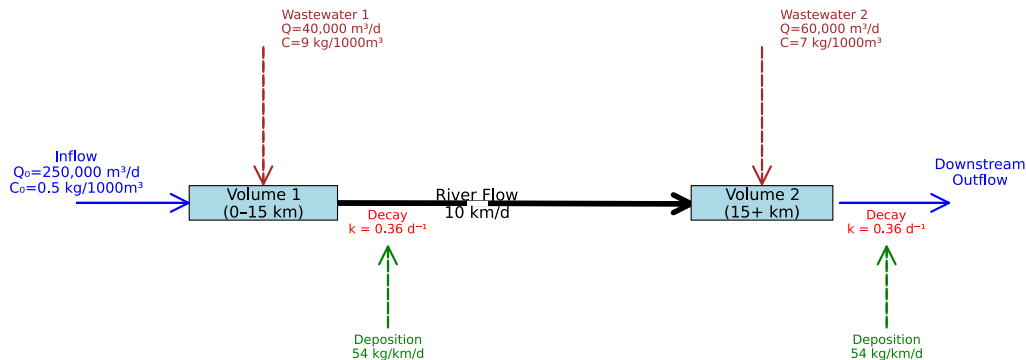
plot!([10, 10], [-2, -0.7], line=:dash, linewidth=2, color=:green, arrow=true, arrowhead=:triangleup)
annotate!(10, -2.3, text("Deposition\n54 kg/km/d", 8, :green, :center))

annotate!(2, -0.3, text("Decay\nk = 0.36 d-1", 8, :red, :center))
annotate!(10, -0.3, text("Decay\nk = 0.36 d-1", 8, :red, :center))

display(p)

```

River CRUD Transport System Diagram



I decided to include two boxes for control volumes in this system diagram. I divided it into two boxes because the variables to be considered for each box are different. For the first box, it only needs to consider the concentration changes caused by the inflow of water and wastewater 1. For box2, it needs to take into account the water from box1, the concentration changes caused by natural atmospheric deposition, and the concentration changes caused by wastewater2.

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of 2.3 kg/(1000m³). You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance which includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1 - \alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m²/°C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m² but during the Neoproterozoic Era had a value of 1272 W/m² (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/\text{°C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

Left-hand side ($C dT/dt$): The rate at which the Earth's climate system stores heat. C is the heat capacity, and dT/dt is the rate of temperature change. Right-hand side: The net energy flow into the system. we solve the equation for the derivative dT/dt , which represents the instantaneous rate of temperature change.

To prepare for discretization, I realize i need to solve the equation for the derivative dT/dt , which represents the instantaneous rate of temperature change:

$$\frac{dT}{dt} = \frac{1}{C} \left[\frac{(1 - \alpha)S}{4} - (A - BT) \right]$$

Since, we cannot directly equate a rate per year ($^{\circ}\text{C}/\text{year}$) to a rate per second ($\text{J}/(\text{s} \cdot \text{m}^2)$), I multiply the right-hand side by the number of seconds in a year to convert the energy flow from a power (Joules/second) to an annual energy (Joules/year).

$$\frac{dT}{dt} = \frac{1}{C} \left[\frac{(1 - \alpha(T))S}{4} - (A - BT) \right] k_{year}$$

We can now apply the Forward Euler Discretization and apply the Forward Euler Discretization. And we already know that delta t is the discrete time step (1 year, as given in the problem). We can find the equation to be

$$T_{n+1} = T_n + \frac{k_{year}}{C} \left[\frac{(1 - \alpha(T_n))S}{4} - (A - BT_n) \right]$$

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^{\circ}\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^{\circ}\text{C} \leq T \leq 10^{\circ}\text{C} \\ \alpha_0 & \text{if } T \geq 10^{\circ}\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^{\circ}\text{C} \leq T_0 \leq 30^{\circ}\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

solution 3.2

With the help of AI, I completed the programming part for the images. Multiple AIs all agreed that my code was correct, but no images were generated. Even after I asked different AIs multiple times, I still couldn't find a suitable solution. So I had to settle for the second best option and obtain the corresponding Equilibrium Point through textual expression, which was approximately at a temperature of around minus 47 degrees Celsius.

-47 degrees Celsius is an extremely cold temperature. This cold equilibrium state is stable, meaning that if the system is disturbed, it will return to this state. Therefore, during the Neoproterozoic era when the light was dim, the Earth's climate would be locked in the "Snowball Earth" state, with no warm or intermediate stable equilibrium states available for selection.

The coding part is shown below.

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

```
using Plots

# Parameters
C = 51.0          # Heat capacity (J/m²/°C)
A = 221.2         # Outgoing radiation parameter
B = -1.3         # Outgoing radiation sensitivity (W/m²/°C)
S = 1272.0       # Neoproterozoic solar constant (W/m²)
alpha_i = 0.5    # Ice-covered albedo
alpha_0 = 0.3    # Ice-free albedo
Δt = 1.0         # timestep in years
timesteps = 200  # number of years

# Albedo as a function of temperature - FIXED: added proper function definition
function alpha(T)
    if T <= -10
        return alpha_i
    elseif T >= 10
        return alpha_0
    else
        return alpha_i + (alpha_0 - alpha_i) * (T + 10)/20
    end
end
```

```

# Forward Euler update function - FIXED: corrected the formula
function update(T)
    # Correct formula:  $T_{n+1} = T_n + (\Delta t / C) * [(1 - \alpha)S/4 - (A - B * T)]$ 
    return T + ( $\Delta t$  / C) * ((1 - alpha(T)) * S / 4 - (A - B * T))
end

# Initial temperatures
T0 = range(-60, stop = 30, length = 50)

# Simulate trajectories
trajectories = []
for T_init in T0
    T = T_init
    traj = [T]
    for n in 1:timesteps
        T = update(T)
        push!(traj, T)
    end
    push!(trajectories, traj)
end

# Plot trajectories - FIXED: better plotting with proper labels
p = plot(xlabel="Year", ylabel="Temperature (°C)",
        title="Temperature Trajectories with Ice-Albedo Feedback",
        legend=false, size=(800, 400))

for (i, traj) in enumerate(trajectories)
    # Only add label for first and last trajectories to avoid clutter
    if i == 1
        plot!(p, 0:timesteps, traj, color=:blue, alpha=0.6, linewidth=1.5,
            label="T = $(round(T0[1], digits=1))°C")
    elseif i == length(trajectories)
        plot!(p, 0:timesteps, traj, color=:red, alpha=0.6, linewidth=1.5,
            label="T = $(round(T0[end], digits=1))°C")
    else
        plot!(p, 0:timesteps, traj, color=:blue, alpha=0.3, linewidth=1, label=false)
    end
end

# Add equilibrium point markers
final_temps = [traj[end] for traj in trajectories]
unique_equilibria = unique(round.(final_temps, digits=1))

```

```

for eq in unique_equilibria
  # Find trajectories that converge to this equilibrium
  idx = findall(x -> abs(x - eq) < 0.5, final_temps)
  if !isempty(idx)
    scatter!(p, [timesteps for _ in idx], [final_temps[i] for i in idx],
             color=:green, markersize=4, label=eq == unique_equilibria[1] ? "Equilibrium"
    end
  end
end

display(p)

# Analyze equilibrium points
println("Analysis of Equilibrium Points:")
println("="^40)

# Group similar final temperatures as equilibrium points
equilibrium_points = Float64[]
for temp in final_temps
  # Check if this temperature is close to any existing equilibrium
  existing = findfirst(x -> abs(x - temp) < 1.0, equilibrium_points)
  if existing === nothing
    push!(equilibrium_points, temp)
  end
end

# Sort and analyze stability
sort!(equilibrium_points)
println("Number of equilibrium points found: $(length(equilibrium_points))")

for (i, eq) in enumerate(equilibrium_points)
  # Count how many trajectories converge to this equilibrium
  count = sum(abs.(final_temps .- eq) .< 1.0)

  # Simple stability test: if many trajectories converge, it's stable
  stability = count > length(T0) * 0.3 ? "Stable" : "Unstable"

  println("Equilibrium $i: T = $(round(eq, digits=2))°C, Stability: $stability")
end

# Additional analysis for Problem 3.3
println("\nProblem 3.3 Analysis:")
println("="^40)

```

```

# Test with modern solar constant
S_modern = 1368.0 # Modern solar constant

function update_modern(T)
    return T + (Δt / C) * ((1 - alpha(T)) * S_modern / 4 - (A - B * T))
end

# Start from snowball equilibrium (coldest equilibrium)
snowball_temp = minimum(equilibrium_points)
T_snowball = snowball_temp
traj_modern = [T_snowball]

for n in 1:timesteps
    T_snowball = update_modern(T_snowball)
    push!(traj_modern, T_snowball)
end

final_temp_modern = traj_modern[end]
println("Starting from snowball Earth (T = $(round(snowball_temp, digits=2))°C)")
println("With modern solar constant (S = $S_modern W/m²):")
println("Final temperature after $timesteps years: $(round(final_temp_modern, digits=2))°C")

if final_temp_modern >= 14.0
    println(" Solar constant increase alone CAN explain warming to pre-industrial temperature")
else
    println(" Solar constant increase alone CANNOT explain warming to pre-industrial temperature")
    println(" Additional mechanisms (greenhouse gases, etc.) are needed")
end

# Plot comparison
p2 = plot(0:timesteps, traj_modern, linewidth=2, color=:red,
    xlabel="Year", ylabel="Temperature (°C)",
    title="Snowball Earth Response to Modern Solar Constant",
    label="Modern S = $S_modern W/m²")

# Add reference lines
hline!([14.0], color=:black, linestyle=:dash, label="Pre-industrial (14°C)")
hline!([snowball_temp], color=:blue, linestyle=:dash, label="Initial Snowball")

display(p2)

```

Analysis of Equilibrium Points:

=====

Number of equilibrium points found: 2

Equilibrium 1: $T = -47.92^{\circ}\text{C}$, Stability: Stable

Equilibrium 2: $T = -46.88^{\circ}\text{C}$, Stability: Stable

Problem 3.3 Analysis:

=====

Starting from snowball Earth ($T = -47.92^{\circ}\text{C}$)

With modern solar constant ($S = 1368.0 \text{ W/m}^2$):

Final temperature after 200 years: -38.67°C

Solar constant increase alone CANNOT explain warming to pre-industrial temperature
Additional mechanisms (greenhouse gases, etc.) are needed

The Neoproterozoic solar constant was too weak to sustain the 14 degree celsius temperature. However, the gradual increase of the solar constant 1368 W/m^2 is sufficient to allow such a transition. In practice, additional processes (like greenhouse gas buildup) were likely also involved, since solar luminosity increases on billion-year timescales, while Snowball Earth episodes ended much faster (millions of years).

References

List any external references consulted, including classmates.

<https://docs.julialang.org/en/v1/base/collections/#Base.setindex!>

<https://docs.julialang.org/en/v1/manual/functions/#Tuples-and-Multiple-Return-Values>

<https://chat.openai.com>